

Important Bootstrap Rules & Conventions

Practical rules to remember when writing Bootstrap HTML for Web Designing Lab

1. Include the Bootstrap CSS inside

The Bootstrap CSS CDN should normally be placed inside the <head> section. This allows Bootstrap styles to be loaded and applied to the page.

```
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">
```

Rule: CSS links go inside <head>.

2. Put the Bootstrap JavaScript near the end of

Yes. This is a common and recommended placement. Put the Bootstrap JavaScript bundle near the end of the <body>, usually just before </body>.

```
<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"></script>
```

Why? The browser can load and display the page content first, while the JavaScript is loaded afterward. It also keeps scripts from unnecessarily delaying the initial rendering of the page.

3. The script tag is not compulsory for every Bootstrap page

If you are only using Bootstrap CSS classes such as buttons, cards, grids, typography, spacing, tables, and basic forms, you do not need the Bootstrap JavaScript bundle.

You need the Bootstrap JavaScript bundle for JavaScript-powered components such as **Carousel**, **Modal**, **Dropdown**, **Offcanvas**, **Accordion**, **Collapse**, **Tooltips**, **Popovers**, and responsive navbar toggling.

4. Use the viewport meta tag

For responsive Bootstrap pages, include the viewport meta tag inside <head>.

```
<meta name="viewport" content="width=device-width, initial-scale=1">
```

Rule: Bootstrap is designed for responsive layouts, so the viewport tag is important for proper display on phones and tablets.

5. Use Bootstrap classes instead of writing unnecessary CSS

Bootstrap provides ready-made classes for many common tasks. For example, use text-center, mt-3, p-4, bg-primary, and btn btn-primary when they meet your requirement.

```
<div class="text-center mt-3 p-4 bg-light">Hello Bootstrap</div>
```

You can still write your own CSS when Bootstrap does not provide the exact design you need.

6. Follow the Bootstrap container and grid structure

For layouts, a common Bootstrap structure is:

```
<div class="container">
<div class="row">
<div class="col-md-6">Column 1</div>
```

```
<div class="col-md-6">Column 2</div>
</div>
</div>
```

Remember: container contains the layout, row creates a row, and col-* creates columns.

7. Correct order of Bootstrap links

A clean Bootstrap HTML page usually follows this order:

Order	What to include	Location
1	DOCTYPE and HTML structure	Top of document
2	Viewport meta tag	<head>
3	Bootstrap CSS CDN	<head>
4	Your own CSS	<head>, usually after Bootstrap CSS
5	Page HTML / Bootstrap components	<body>
6	Bootstrap JS bundle	End of <body>
7	Your own JavaScript	End of <body>, as appropriate

8. If using your own CSS, load it after Bootstrap CSS

This is an important practical rule. If your custom CSS needs to override Bootstrap's styles, placing your CSS after Bootstrap makes the intended override easier.

```
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">
<link rel="stylesheet" href="style.css">
```

Bootstrap first, your custom CSS second.

9. Use Bootstrap's correct class names

Bootstrap classes are predefined. A small spelling mistake means the class will not work.

```
<button class="btn btn-primary">Correct</button>
<button class="btn btn-danger">Delete</button>
```

For example, use `btn btn-primary`, not an invented class such as `button-blue`, unless you have created that custom class yourself.

10. Bootstrap JavaScript bundle is usually the easiest option

For Bootstrap 5, the `bootstrap.bundle.min.js` file is convenient because it includes Bootstrap's required JavaScript functionality and Popper where needed.

For normal lab work, using the bundle CDN is simpler than manually adding separate JavaScript dependencies.

11. Do not confuse Bootstrap CSS with Bootstrap JavaScript

Bootstrap CSS	Bootstrap JavaScript
Provides styling	Provides interactive behavior
Buttons and colors	Modal
Grid and layout	Carousel

Bootstrap CSS	Bootstrap JavaScript
Typography and spacing	Dropdown
Cards and forms	Accordion / Collapse

12. A Good Basic Bootstrap HTML Structure

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">
<link rel="stylesheet" href="style.css">
</head>
<body>

<div class="container">
<h1 class="text-center">My Bootstrap Page</h1>
<button class="btn btn-primary">Click Me</button>
</div>

<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

Quick Rules to Remember

- **CSS CDN:** Put it inside `<head>`.
- **Viewport:** Include the responsive viewport meta tag.
- **JS bundle:** If needed, put it near the end of `<body>`.
- **Custom CSS:** Put it after Bootstrap CSS when you want to override Bootstrap styles.
- **JavaScript is optional:** Only include Bootstrap JS when your components need it.
- **Grid:** Remember the common container > row > col structure.
- **Bootstrap 5:** Use Bootstrap 5 syntax and CDN versions consistently.

Most important answer to your question: Yes, placing the Bootstrap JS script near the very end of `<body>`, just before `</body>`, is a common and recommended practice. But the script itself is only required when you use Bootstrap's JavaScript-powered features.